Question 1

Which characteristic most clearly distinguishes Generative AI from traditional AI systems?

A. It only performs mathematical calculations

B. It creates new content by learning patterns from existing data

C. It requires internet access for every task

D. It only analyzes structured databases

Question 2

Which example best represents an AI agent?

A. A static webpage displaying weather information

B. A calculator performing arithmetic operations

C. A cybersecurity system that monitors network traffic, detects threats, and automatically isolates infected devices

D. A spreadsheet sorting data alphabetically

Question 3

Why are multimodal AI systems generally more powerful than single-modal AI systems?

A. They only process larger datasets.

B. They combine multiple types of data to produce richer insights.

C. They eliminate all forms of AI bias.

D. They never require human oversight.

Question 4

Which statement best describes a hallucination in a large language model?

A. The model intentionally deceives users.

B. The model generates plausible but incorrect information.

C. The model accesses unauthorized databases.

D. The model refuses to answer questions.

Question 5

Which scenario is an example of AI safety rather than AI security?

A. Preventing hackers from stealing a model

B. Encrypting AI training datasets

C. Reducing hallucinations in a medical chatbot

D. Preventing ransomware attacks against an AI server

Question 6

Which scenario best illustrates AI security?

A. Improving model explainability

B. Protecting an AI system against prompt injection attacks

C. Reducing demographic bias

D. Improving model fairness

Question 7

Why should users verify information produced by an LLM?

A. LLMs always generate outdated information.

B. LLMs cannot distinguish factual truth from statistically likely text.

C. LLMs intentionally fabricate information.

D. LLMs only work with public internet data.

Question 8

Which application demonstrates Generative AI being integrated into public services?

A. A government chatbot answering citizen questions in multiple languages

B. A paper filing system in a government office

C. A traditional call center with human operators only

D. A website displaying PDF documents without interaction

Question 9

Which technology is primarily responsible for creating highly realistic deepfake images?

A. Decision Trees

B. Diffusion models

C. Linear Regression

D. Rule-based Expert Systems

Question 10

Why do deepfakes present a growing societal risk?

A. They reduce image quality.

B. They increase internet bandwidth requirements.

C. They make fabricated media increasingly difficult to distinguish from authentic evidence.

D. They eliminate the need for photography.

Question 11

Which business benefit is most directly associated with AI-powered customer service chatbots?

A. Eliminating customer complaints

B. Providing personalized assistance around the clock

C. Preventing cyberattacks

D. Removing the need for human employees

Question 12

What is one limitation of many standalone large language models?

A. They cannot process natural language.

B. They are always connected to live internet data.

C. Their knowledge may become outdated unless connected to external information sources.

D. They cannot summarize documents.

Question 13

Which statement best reflects Responsible AI?

A. Maximizing automation regardless of consequences

B. Prioritizing performance over privacy

C. Designing AI systems that are ethical, transparent, accountable, and aligned with human values

D. Preventing competitors from copying AI models

Question 14

A hospital AI accidentally exposes confidential patient information to an unauthorized individual. Why could this incident involve both AI safety and AI security?

A. Safety and security are identical concepts.

B. Safety concerns involve harm to individuals, while security concerns involve unauthorized access.

C. Only security applies because patient records were digital.

D. Only safety applies because AI made the mistake.

Question 15

Which situation best demonstrates the value of multimodal AI in healthcare?

A. Translating medical documents into another language

B. Combining X-rays, physician notes, and laboratory results to support diagnosis

C. Searching medical journals by keyword

D. Scheduling patient appointments

Question 16

Why are AI-generated deepfakes increasingly difficult to detect?

A. Cameras automatically verify authenticity.

B. Neural networks continuously improve image realism.

C. Modern browsers automatically remove fake content.

D. Deepfakes contain visible digital watermarks.

Question 17

What is the primary purpose of algorithmic fairness?

A. Increasing processing speed

B. Eliminating cybersecurity threats

C. Reducing unfair outcomes across different groups

D. Compressing AI models

Question 18

Which statement best describes a chatbot?

A. A database management system

B. A software application designed to simulate conversations with users

C. A programming language

D. A cybersecurity firewall

Question 19

Why is Responsible AI considered a strategic business imperative?

A. It guarantees AI systems will never fail.

B. It builds trust while reducing legal, ethical, and operational risks.

C. It eliminates all regulatory requirements.

D. It completely removes the need for governance.

Question 20

Which statement best summarizes the relationship between AI safety and AI security?

A. They are unrelated disciplines.

B. AI safety focuses on preventing unintentional harm, while AI security focuses on protecting systems from intentional attacks.

C. AI safety only applies to robotics.

D. AI security only applies after deployment.

Answer Key

Q	Ans	Q	Ans
1	B	11	B
2	C	12	C
3	B	13	C
4	B	14	B
5	C	15	B
6	B	16	B
7	B	17	C
8	A	18	B
9	B	19	B

Q	Ans	Q	Ans
10	C	20	B

Explanations

Q1 – B
Generative AI differs from traditional AI because it **creates new content** (text, images, audio, code, etc.) by learning patterns from existing data.

Q2 – C
AI agents observe their environment, make decisions, and act autonomously. Automatically isolating infected devices demonstrates all three capabilities.

Q3 – B
Multimodal AI combines information from multiple data types (text, images, audio, etc.), enabling richer understanding and better decision-making.

Q4 – B
Hallucinations occur when an LLM confidently generates information that is incorrect or unsupported by facts.

Q5 – C
Reducing hallucinations addresses unintended harmful behavior, which is an AI safety concern.

Q6 – B
Prompt injection is an intentional attack designed to manipulate an AI system, making it an AI security issue.

Q7 – B
LLMs generate text based on learned statistical patterns rather than verifying factual truth, so outputs should always be reviewed.

Q8 – A
Government chatbots are a practical example of generative AI improving citizen services and accessibility.

Q9 – B
Modern deepfakes are commonly produced using diffusion-based image generation models capable of creating highly realistic synthetic media.

Q10 – C
Deepfakes threaten trust because fabricated media can become nearly indistinguishable from authentic evidence.

Q11 – B
AI chatbots improve customer experience by providing personalized assistance at any time of day.

Q12 – C
Many LLMs rely on knowledge learned during training and may not contain current information unless integrated with external retrieval or search systems.

Q13 – C
Responsible AI emphasizes ethics, transparency, accountability, fairness, privacy, and alignment with human values throughout the AI lifecycle.

Q14 – B
The disclosure harms patients (safety) while also exposing confidential information to unauthorized individuals (security).

Q15 – B
Healthcare multimodal AI combines different sources of clinical information to improve diagnosis and treatment decisions.

Q16 – B
Advances in deep learning and diffusion models have made synthetic media increasingly realistic and difficult to detect.

Q17 – C
Algorithmic fairness seeks to reduce discriminatory outcomes and improve equitable treatment across different populations.

Q18 – B
A chatbot is software designed to simulate human conversation using text or voice interactions.

Q19 – B
Responsible AI strengthens stakeholder trust, improves compliance, and reduces ethical, legal, and operational risks.

Q20 – B
AI safety focuses on preventing unintended harm caused by AI behavior, while AI security focuses on defending AI systems against intentional attacks.

Questions

Question 21

A company deploys a generative AI tool that provides different answers to the same question on different occasions. Which characteristic of many LLMs best explains this behavior?

A. They intentionally randomize responses to confuse users.

B. They generate responses probabilistically based on learned language patterns.

C. They continuously retrain themselves using user conversations.

D. They retrieve different documents from the internet every time.

Question 22

Which ethical concern is most closely associated with the environmental impact of large language models?

A. Data poisoning

B. High computational energy consumption

C. Identity theft

D. Prompt injection

Question 23

Why does the Module state that a generic AI ethics checklist is insufficient?

A. AI systems do not require ethical reviews.

B. Every AI application presents context-specific risks that require individual evaluation.

C. Ethics only applies to healthcare AI.

D. Ethics checklists are prohibited by regulators.

Question 24

Which example best demonstrates AI undermining truth and truth-telling?

A. An AI system compresses image files.

B. A chatbot confidently provides fabricated legal advice.

C. A recommendation engine suggests movies.

D. A search engine indexes new websites.

Question 25

Which ethical risk is associated with AI systems expressing emotions such as "I understand how you feel"?

A. Evaluation bias

B. Artificial empathy

C. Historical bias

D. Deployment bias

Question 26

Why can artificial empathy become harmful?

A. AI systems actually experience emotions.

B. Users may develop emotional dependence or be manipulated.

C. AI becomes conscious.

D. AI stops generating responses.

Question 27

Which scenario represents a privacy concern highlighted in Module 3?

A. AI automatically summarizes meeting notes.

B. A chatbot stores personal conversations indefinitely to improve future models.

C. AI translates documents into multiple languages.

D. AI generates software code.

Question 28

Which question best illustrates the authorship challenge of Generative AI?

A. How much RAM does the model require?

B. Who owns AI-generated artwork?

C. How many GPUs were used?

D. What programming language built the model?

Question 29

An organization discovers its AI writing assistant occasionally copies an author's writing style almost perfectly. Which ethical issue is most relevant?

A. Network security

B. Copyright and intellectual property

C. Cloud scalability

D. Data compression

Question 30

Why is explainability important in Responsible AI?

A. It allows users to understand how AI reached its conclusions.

B. It improves processor speed.

C. It guarantees perfect accuracy.

D. It removes the need for governance.

Question 31

A facial recognition model performs significantly better on one demographic group than another because that group dominated the training data. Which type of bias is MOST directly responsible?

A. Representation bias

B. Deployment bias

C. Evaluation bias

D. Prompt bias

Question 32

Which situation best illustrates historical bias?

A. A hiring model learns from decades of hiring records dominated by male applicants.

B. A model is deployed in another country.

C. A benchmark dataset contains incorrect labels.

D. A user enters misleading prompts.

Question 33

A healthcare AI system performs poorly for minority populations because those populations were underrepresented during data collection. Which bias does this demonstrate?

A. Deployment bias

B. Representation bias

C. Optimization bias

D. Automation bias

Question 34

Which mitigation strategy would MOST effectively reduce representation bias before model training?

A. Increase model size.

B. Use a more powerful GPU.

C. Rebalance the dataset to better represent the target population.

D. Reduce the number of training examples.

Question 35

Evaluation bias occurs when:

A. The model is tested against inappropriate benchmarks.

B. The dataset contains duplicate records.

C. Users intentionally attack the AI.

D. The model has insufficient computing power.

Question 36

A clothing recommendation model trained on purchasing behavior of young adults in the United States is deployed for elderly customers in another country. Which bias is illustrated?

A. Historical bias

B. Deployment bias

C. Representation bias

D. Confirmation bias

Question 37

Which statement best describes algorithmic bias?

A. Bias introduced solely through poor data collection.

B. Bias arising from algorithm design choices despite balanced data.

C. Bias caused only by malicious attackers.

D. Bias occurring after software installation.

Question 38

Amazon discontinued its AI recruiting tool primarily because:

A. It processed applications too slowly.

B. It learned discriminatory hiring patterns from historical resumes.

C. It could not parse PDF documents.

D. It generated interview questions incorrectly.

Question 39

Why did critics argue that the Apple Card algorithm could still produce gender bias even if gender was not explicitly included?

A. Gender is impossible to remove from databases.

B. Proxy variables can indirectly reflect protected characteristics.

C. Credit scoring always requires gender.

D. The algorithm used facial recognition.

Question 40

What is the primary lesson shared by the Amazon hiring tool, facial recognition, and Apple Card case studies?

A. AI systems are always objective.

B. Bias can emerge throughout the AI lifecycle and requires continuous governance.

C. Larger AI models eliminate fairness concerns.

D. AI should not be used in business.

Answer Key

Q	Ans	Q	Ans
21	B	31	A
22	B	32	A
23	B	33	B
24	B	34	C
25	B	35	A
26	B	36	B
27	B	37	B
28	B	38	B
29	B	39	B
30	A	40	B

Explanations

Q21 — B

LLMs generate responses probabilistically using learned language patterns. This stochastic behavior means identical prompts can produce different outputs.

Q22 — B

Training and operating large language models require significant computational resources and electricity, creating environmental concerns.

Q23 — B

Ethical risks vary by application. Healthcare, finance, education, and cybersecurity all have different risks, so a universal checklist alone is insufficient.

Q24 — B

Hallucinations can produce convincing but false information, undermining trust and truthful decision-making.

Q25 — B

Artificial empathy refers to AI simulating emotional understanding despite having no genuine emotions.

Q26 — B

Users may trust emotionally expressive AI too much, creating emotional dependence or opportunities for manipulation.

Q27 — B

Indefinite storage of personal conversations creates significant privacy and data protection concerns.

Q28 — B

One major legal and ethical question is ownership of AI-generated content and responsibility for copyright.

Q29 — B

Closely mimicking an artist's style raises concerns about copyright, intellectual property, and authorship.

Q30 — A

Explainability helps users understand why AI made a particular decision, improving transparency and accountability.

Q31 — A

Representation bias occurs when training data does not adequately represent the population the model will serve.

Q32 — A

Historical bias reflects existing societal patterns embedded in historical training data.

Q33 — B

Underrepresentation of certain populations produces representation bias and often poorer model performance for those groups.

Q34 — C

Rebalancing datasets and improving diversity before training are common techniques for reducing representation bias.

Q35 — A

Evaluation bias occurs when inappropriate or non-representative benchmarks are used to assess model performance.

Q36 — B

Using a model outside the environment for which it was designed is a classic example of deployment bias.

Q37 — B

Algorithmic bias can arise from optimization choices, model architecture, or training methods even when the dataset itself has been carefully balanced.

Q38 — B

Amazon's recruiting model learned historical hiring preferences reflected in past resumes, leading to discrimination against women.

Q39 — B

Variables such as spending patterns or credit history can indirectly encode gender-related information, creating proxy bias.

Q40 — B

All three industry examples demonstrate that bias can originate from data, algorithms, evaluation, or deployment, reinforcing the importance of Responsible AI practices throughout the AI lifecycle.

Questions

Question 41

An organization is defining the business objective for an AI system while identifying potential harms, affected stakeholders, and ethical concerns before collecting any data. Which stage of the Responsible AI Lifecycle is this?

A. Model Development

B. Problem Framing

C. Deployment

D. Monitoring and Feedback

Question 42

Which activity belongs primarily to the Data Collection stage of the Responsible AI Lifecycle?

A. Monitoring model drift

B. Ensuring data is representative and privacy compliant

C. Fine-tuning model hyperparameters

D. Conducting penetration testing

Question 43

During model development, which action best supports Responsible AI?

A. Ignoring demographic performance differences

B. Applying fairness techniques and testing model robustness

C. Maximizing accuracy regardless of bias

D. Eliminating human oversight

Question 44

Which activity is most appropriate during the Model Testing stage?

A. Collecting new customer data

B. Evaluating model performance against predefined requirements using separate validation data

C. Launching the model into production immediately

D. Removing monitoring controls

Question 45

Why is continuous monitoring essential after deployment?

A. AI models never change after deployment.

B. Real-world conditions and data distributions can evolve over time.

C. Monitoring is only required by government agencies.

D. Monitoring guarantees zero hallucinations.

Question 46

Which stage specifically focuses on incorporating stakeholder feedback and retraining the model when necessary?

A. Data Collection

B. Monitoring and Feedback

C. Problem Framing

D. Model Development

Question 47

A hospital AI system analyzes CT scans, physician notes, laboratory results, and genomic information simultaneously. What type of AI system is this?

A. Rule-based Expert System

B. Large Language Model

C. Multimodal AI System

D. Robotic Process Automation

Question 48

What is the primary advantage of multimodal AI in healthcare?

A. It reduces electricity consumption.

B. It combines multiple information sources to improve diagnosis and treatment decisions.

C. It completely replaces physicians.

D. It eliminates patient consent requirements.

Question 49

MSK-MIND was developed primarily to address which limitation?

A. Slow internet connectivity

B. Inability to integrate diverse healthcare data sources

C. Shortage of cloud storage

D. High cost of medical imaging equipment

Question 50

Which combination of data would BEST demonstrate multimodal AI?

A. Two different spreadsheets

B. MRI images, physician notes, and laboratory results

C. Customer names and addresses

D. Two versions of the same document

Question 51

According to the mental health chatbot case study, these tools should primarily be viewed as:

A. Complete replacements for psychiatrists

B. First-line support tools that require appropriate human oversight

C. Emergency crisis intervention systems only

D. Diagnostic systems with legal authority

Question 52

Which technology allows mental health chatbots to understand emotional context?

A. Blockchain

B. Natural Language Processing (NLP)

C. Quantum Computing

D. Optical Character Recognition

Question 53

Why is crisis detection an important capability in mental health chatbots?

A. It reduces electricity costs.

B. It helps identify high-risk situations requiring escalation to human professionals.

C. It improves image generation.

D. It increases chatbot response speed.

Question 54

Which limitation was commonly reported by users of AI mental health chatbots?

A. Lack of internet connectivity

B. Process violations such as repetitive or misunderstood conversations

C. Excessive computational power

D. Poor handwriting recognition

Question 55

Which statement best reflects Responsible AI use of mental health chatbots?

A. Users should rely exclusively on AI during psychological crises.

B. AI chatbots should complement—not replace—qualified mental health professionals.

C. Human oversight reduces AI effectiveness.

D. Chatbots are suitable for every mental health condition.

Question 56

Charlotte AI was created primarily to help organizations address:

A. Database optimization

B. The shortage of experienced cybersecurity analysts

C. Mobile application development

D. Website design automation

Question 57

What architectural feature distinguishes Charlotte AI?

A. A single rule-based expert system

B. Multiple specialized AI agents working together

C. Only traditional machine learning algorithms

D. Manual threat analysis only

Question 58

How does Charlotte AI reduce hallucinations?

A. By preventing users from asking questions

B. Through dedicated validation agents and layered verification

C. By limiting responses to one sentence

D. By disconnecting from cloud infrastructure

Question 59

Why are false negatives generally considered more dangerous than false positives in cybersecurity?

A. False negatives consume more memory.

B. False negatives allow genuine attacks to remain undetected.

C. False negatives are easier to investigate.

D. False negatives improve detection accuracy.

Question 60

A financial institution deploys an AI system and regularly measures fairness, accuracy, user complaints, and performance degradation. Which Responsible AI principle is being demonstrated?

A. Historical bias

B. Continuous monitoring and governance

C. Data minimization only

D. Prompt engineering

Answer Key

Q	Ans	Q	Ans
41	B	51	B
42	B	52	B

Q	Ans	Q	Ans
43	B	53	B
44	B	54	B
45	B	55	B
46	B	56	B
47	C	57	B
48	B	58	B
49	B	59	B
50	B	60	B

Explanations

Q41 — B

Problem Framing is the first stage of the Responsible AI Lifecycle. It defines the business problem, identifies stakeholders, and considers ethical risks before development begins.

Q42 — B

Representative, unbiased, and privacy-compliant data is foundational to responsible AI development and helps reduce downstream fairness issues.

Q43 — B

Responsible AI requires fairness testing, documentation, robustness testing, and safety evaluations during model development—not just maximizing predictive accuracy.

Q44 — B

Model testing validates whether the system satisfies technical, business, safety, and fairness requirements using data separate from the training dataset.

Q45 — B

Data distributions, user behavior, regulations, and threats evolve after deployment. Continuous monitoring helps detect model drift, emerging bias, and performance degradation.

Q46 — B

Monitoring and Feedback is the lifecycle phase where organizations continuously evaluate outputs, collect stakeholder feedback, and retrain models when appropriate.

Q47 — C

Multimodal AI integrates multiple data modalities (images, text, laboratory values, genomics, etc.) into a unified analysis.

Q48 — B

By combining complementary information from several data sources, multimodal AI provides richer context for diagnosis, prognosis, and treatment planning.

Q49 — B

MSK-MIND was designed to overcome fragmented healthcare datasets by integrating radiology, pathology, genomics, clinical records, and patient-reported outcomes into a single research platform.

Q50 — B

Using different modalities—medical images, clinical notes, and laboratory results—is the defining characteristic of multimodal AI.

Q51 — B

Mental health chatbots improve access to support but should supplement—not replace—licensed mental health professionals, especially during crises.

Q52 — B

Natural Language Processing (NLP) enables chatbots to interpret user language, identify emotional cues, and generate contextually appropriate responses.

Q53 — B

Crisis detection identifies situations such as suicidal ideation so users can be escalated promptly to human intervention.

Q54 — B

Users most frequently reported repetitive conversations, misunderstanding of responses, and other process violations caused by NLP limitations.

Q55 — B

Responsible deployment recognizes both the benefits and limitations of AI mental health tools, emphasizing ongoing human oversight.

Q56 — B

Charlotte AI was introduced to help Security Operations Centers (SOCs) cope with increasing alert volumes and the shortage of experienced cybersecurity professionals.

Q57 — B

Charlotte AI employs an agentic architecture consisting of multiple specialized AI agents responsible for functions such as retrieval, threat analysis, and validation.

Q58 — B

Rather than relying solely on one LLM, Charlotte AI uses layered validation and dedicated validation agents to reduce hallucinations and improve trustworthiness.

Q59 — B

False negatives represent genuine attacks that go undetected, potentially leading to successful security breaches, making them generally more serious than false positives.

Q60 — B

Responsible AI extends beyond deployment. Continuous monitoring, governance, fairness assessment, and performance evaluation are necessary throughout the operational life of an AI system.

Question 61

A hospital deploys an AI diagnostic assistant. During testing, auditors discover that diagnostic accuracy for one ethnic group is significantly lower than for others. What should be the team's FIRST priority?

A. Increase GPU capacity

B. Investigate potential bias in the training and evaluation datasets

C. Deploy the model immediately and fix issues later

D. Increase model complexity

Question 62

A bank intentionally excludes gender from its credit-scoring model. However, women consistently receive lower credit limits due to correlated spending behavior. What type of bias is MOST likely occurring?

A. Historical bias

B. Proxy data bias

C. Evaluation bias

D. Deployment bias

Question 63

An AI recruiting tool consistently favors applicants from prestigious universities because historical hiring data contained that pattern. Which Responsible AI principle was MOST likely overlooked?

A. Fairness evaluation during model development

B. Encryption of training data

C. GPU optimization

D. API authentication

Question 64

A company deploys a customer service chatbot that frequently invents refund policies not found in company documentation. Which control would MOST effectively reduce this risk?

A. Larger training datasets only

B. Retrieval-Augmented Generation (RAG) using official company documentation

C. Faster internet connectivity

D. Higher-resolution user interfaces

Question 65

An AI-generated video falsely shows a CEO announcing fraudulent financial results. Which societal risk is MOST directly demonstrated?

A. Data compression

B. Content authenticity and misinformation

C. Software licensing

D. Model optimization

Question 66

A security operations center uses an AI assistant to summarize security alerts. Analysts are required to verify every recommendation before taking action. Which Responsible AI principle is being applied?

A. Human oversight

B. Historical bias

C. Deployment bias

D. Artificial empathy

Question 67

Which scenario BEST illustrates AI safety rather than AI security?

A. Preventing prompt injection attacks

B. Ensuring an autonomous medical AI does not recommend harmful treatments

C. Blocking unauthorized API access

D. Encrypting inference requests

Question 68

Which activity would MOST improve transparency in an AI loan approval system?

A. Increasing model size

B. Documenting factors influencing decisions and providing explanations to applicants

C. Adding more hidden layers

D. Encrypting training data

Question 69

A company trains an AI model using customer data collected without obtaining proper consent. Which Responsible AI principle has been violated?

A. Privacy and data governance

B. Model optimization

C. Computational efficiency

D. Prompt engineering

Question 70

An AI medical assistant analyzes X-rays, blood test results, physician notes, and wearable device data simultaneously. What is the PRIMARY benefit?

A. Reduced storage costs

B. Better clinical decision-making through richer contextual understanding

C. Elimination of medical professionals

D. Faster internet performance

Question 71

Which practice BEST reduces historical bias before model training?

A. Collecting a more representative dataset and involving diverse stakeholders

B. Increasing processor speed

C. Deploying the model sooner

D. Reducing the training dataset size

Question 72

Which statement BEST explains why fairness reviews should continue after deployment?

A. Bias can emerge as populations and data distributions change.

B. Fairness only matters before deployment.

C. Models cannot drift after release.

D. Monitoring reduces electricity consumption.

Question 73

A mental health chatbot detects language indicating possible self-harm. What is the MOST appropriate action?

A. Continue automated conversation indefinitely

B. Escalate to appropriate human intervention according to predefined procedures

C. Ignore the message to avoid liability

D. Generate motivational quotes only

Question 74

Charlotte AI uses multiple specialized AI agents instead of one large model primarily because:

A. Agent specialization improves efficiency, validation, and task performance.

B. GPUs cannot support one model.

C. Regulations prohibit single-model architectures.

D. It eliminates hallucinations completely.

Question 75

Which statement BEST describes the relationship between explainability and accountability?

A. Explainability helps organizations demonstrate accountability for AI decisions.

B. Explainability removes legal responsibility.

C. Accountability eliminates hallucinations.

D. They are unrelated concepts.

Question 76

An AI model accurately predicts disease risk but cannot explain how it reached its conclusions. Which Responsible AI challenge does this represent?

A. Transparency and explainability

B. Deployment bias

C. Historical bias

D. Cybersecurity

Question 77

An organization performs periodic audits comparing model accuracy across age groups, genders, and ethnicities. What is the PRIMARY objective?

A. Reduce electricity usage

B. Detect emerging fairness issues

C. Increase processing speed

D. Reduce storage requirements

Question 78

Which scenario BEST demonstrates deployment bias?

A. A fraud detection model trained on one country's banking behavior performs poorly when deployed in another country without adaptation.

B. A balanced dataset is collected.

C. A model undergoes fairness testing.

D. A chatbot answers customer questions.

Question 79

Which governance practice would MOST effectively build user trust in Generative AI systems?

A. Hiding model limitations

B. Clearly communicating capabilities, limitations, and sources of uncertainty

C. Eliminating documentation

D. Preventing users from asking follow-up questions

Question 80

A Responsible AI review board discovers that an AI model performs well technically but disadvantages a protected demographic. What should be the MOST appropriate decision?

A. Deploy because technical accuracy is sufficient.

B. Revisit data, algorithms, and fairness controls before deployment.

C. Ignore fairness because bias is unavoidable.

D. Replace the model with manual spreadsheets.

Answer Key

Q	**Ans**	**Q**	**Ans**
61	B	71	A

Q	Ans	Q	Ans
62	B	72	A
63	A	73	B
64	B	74	A
65	B	75	A
66	A	76	A
67	B	77	B
68	B	78	A
69	A	79	B
70	B	80	B

Explanations

Q61 — B

When significant performance disparities appear across demographic groups, the first step is to investigate dataset quality, representation, and evaluation methods before deployment.

Q62 — B

Even when protected attributes like gender are excluded, correlated variables can indirectly encode them, resulting in **proxy data bias**.

Q63 — A

Responsible AI requires evaluating fairness during model development. Historical hiring data should not be accepted without bias assessment and mitigation.

Q64 — B

Retrieval-Augmented Generation (RAG) grounds responses in verified organizational knowledge, significantly reducing hallucinations.

Q65 — B

Deepfakes threaten public trust by enabling convincing misinformation and fabricated evidence.

Q66 — A

Keeping humans responsible for reviewing AI recommendations is a key Responsible AI control known as **human oversight**.

Q67 — B

Preventing harmful recommendations is an AI safety objective because it focuses on avoiding unintended harm rather than defending against attacks.

Q68 — B

Providing understandable reasons for AI decisions improves transparency, explainability, and user trust.

Q69 — A

Collecting personal data without informed consent violates privacy principles and responsible data governance.

Q70 — B

Multimodal AI synthesizes multiple healthcare data types, allowing more comprehensive and context-aware clinical decisions.

Q71 — A

Representative datasets and diverse development teams are effective ways to reduce historical bias before training.

Q72 — A

Fairness is not static. New users, changing demographics, and evolving data can introduce new forms of bias after deployment.

Q73 — B

Mental health chatbots should escalate high-risk situations, such as indications of self-harm, to qualified human professionals.

Q74 — A

Charlotte AI distributes specialized tasks across multiple AI agents, improving efficiency, modularity, and validation.

Q75 — A

Explainability enables organizations to justify AI decisions, supporting accountability and regulatory compliance.

Q76 — A

An accurate but unexplainable model raises concerns about transparency and explainability, especially in high-risk domains.

Q77 — B

Comparing model performance across demographic groups helps identify fairness issues that may require mitigation.

Q78 — A

Applying a model to a different population or environment than the one it was designed for is a classic example of deployment bias.

Q79 — B

Openly communicating an AI system's strengths, weaknesses, limitations, and uncertainty is essential for building trust.

Q80 — B

Technical performance alone is insufficient. Responsible AI requires addressing fairness issues before deployment to reduce harm and improve trustworthiness.

Questions

Question 81

A government agency plans to deploy an AI chatbot to answer citizen questions about taxes and benefits. Which action should be completed **before deployment**?

A. Increase GPU memory

B. Evaluate accuracy, fairness, privacy, and security risks

C. Disable user feedback

D. Publish the chatbot without testing

Question 82

A company uses an LLM to draft legal contracts. What is the MOST appropriate control?

A. Allow the model to finalize contracts independently

B. Require qualified legal professionals to review AI-generated content

C. Train employees to trust every AI response

D. Prevent users from editing AI-generated text

Question 83

A hospital deploys a multimodal AI system that combines medical images, physician notes, and laboratory results. Which Responsible AI principle is MOST critical due to the sensitivity of the data?

A. Entertainment value

B. Data privacy and informed consent

C. Internet bandwidth optimization

D. Marketing effectiveness

Question 84

Which statement BEST explains why Responsible AI spans the entire AI lifecycle?

A. Ethical risks arise only after deployment.

B. Ethical, legal, and operational risks can emerge during planning, development, deployment, and monitoring.

C. Ethics only applies to data collection.

D. Security replaces Responsible AI after deployment.

Question 85

A deepfake video falsely portrays a political leader making inflammatory statements before an election. What is the PRIMARY societal concern?

A. Increased storage requirements

B. Manipulation of public trust and democratic processes

C. Reduced video quality

D. Slower internet performance

Question 86

An AI model achieves 99% accuracy overall but only 72% accuracy for older adults. What does this indicate?

A. The model is unbiased because overall accuracy is high.

B. Performance should be evaluated across relevant demographic groups.

C. Monitoring is unnecessary.

D. The model should automatically be deployed.

Question 87

A cybersecurity AI blocks thousands of legitimate network activities while successfully detecting every attack. What type of issue is MOST likely occurring?

A. Excessive false positives

B. Excessive false negatives

C. Historical bias

D. Data poisoning

Question 88

Which practice BEST improves user trust when interacting with Generative AI?

A. Claim the AI is always correct.

B. Clearly disclose that responses may contain errors and require verification.

C. Hide system limitations.

D. Prevent users from questioning outputs.

Question 89

An organization retrains its AI model every six months using new operational data after validating fairness and performance. Which Responsible AI practice does this demonstrate?

A. Continuous improvement through monitoring and governance

B. Historical bias

C. Deployment bias

D. Artificial empathy

Question 90

A company discovers its AI hiring model consistently rejects qualified candidates from one region because the historical training data contained very few successful applicants from that region. What is the BEST corrective action?

A. Continue using the model because historical data is objective.

B. Improve dataset representation and reassess model fairness before redeployment.

C. Increase processor speed.

D. Remove all monitoring.

Question 91

Which scenario BEST demonstrates the difference between AI Safety and AI Security?

A. Preventing harmful medical recommendations (Safety) and defending against prompt injection attacks (Security)

B. Encrypting databases (Safety) and reducing hallucinations (Security)

C. Improving explainability (Security) and using firewalls (Safety)

D. Collecting representative data (Security) and preventing malware (Safety)

Question 92

A healthcare AI assistant recommends a treatment that contradicts current medical guidelines because its training data is outdated. Which limitation is MOST directly illustrated?

A. Hallucination and outdated knowledge

B. Prompt injection

C. Data encryption failure

D. Network latency

Question 93

Which governance mechanism is MOST effective for overseeing high-risk AI systems?

A. Eliminating documentation requirements

B. Establishing an AI governance committee with defined accountability

C. Removing human review

D. Allowing developers to self-certify models

Question 94

A mental health chatbot detects escalating emotional distress but continues the conversation without escalation. Which Responsible AI principle has MOST likely failed?

A. Human oversight and safety controls

B. Explainability

C. Model compression

D. Energy efficiency

Question 95

Why should organizations evaluate AI systems using multiple metrics instead of accuracy alone?

A. Accuracy captures every important Responsible AI concern.

B. Fairness, robustness, privacy, explainability, and safety are also essential.

C. Accuracy is irrelevant.

D. Only cybersecurity matters.

Question 96

Which statement BEST describes the relationship between Responsible AI and regulatory compliance?

A. Responsible AI replaces regulations.

B. Responsible AI supports compliance but often extends beyond minimum legal requirements.

C. Regulations eliminate ethical responsibilities.

D. Compliance guarantees fairness.

Question 97

A company publishes documentation describing its AI model's intended purpose, known limitations, training assumptions, and appropriate use cases. Which Responsible AI principle is primarily supported?

A. Transparency

B. Historical bias

C. Prompt engineering

D. Model optimization

Question 98

Which of the following represents the BEST example of "human-in-the-loop" governance?

A. AI independently approves all mortgage applications.

B. AI recommends mortgage decisions, but trained analysts review high-risk cases before final approval.

C. AI automatically rejects all incomplete applications.

D. AI continuously retrains itself without supervision.

Question 99

An organization discovers that users are beginning to rely on an AI assistant for decisions outside its intended purpose. What is the MOST appropriate response?

A. Expand the AI's authority immediately.

B. Strengthen user guidance, monitoring, and governance while clarifying intended use.

C. Ignore the behavior because users are responsible.

D. Disable all monitoring.

Question 100

Which statement BEST summarizes the overall objective of Responsible AI?

A. Maximize automation regardless of consequences.

B. Develop and operate AI systems that are ethical, fair, transparent, secure, accountable, and aligned with human values throughout their lifecycle.

C. Focus exclusively on improving model accuracy.

D. Eliminate all human involvement in decision-making.

Answer Key

Q	Ans	Q	Ans
81	B	91	A
82	B	92	A
83	B	93	B
84	B	94	A
85	B	95	B
86	B	96	B
87	A	97	A
88	B	98	B
89	A	99	B
90	B	100	B

Explanations

Q81 — B

Before deployment, organizations should assess accuracy, fairness, privacy, security, and other Responsible AI risks.

Q82 — B

AI-generated legal documents should always be reviewed by qualified legal professionals because LLMs can produce incorrect or incomplete information.

Q83 — B

Multimodal healthcare systems process highly sensitive information, making privacy, consent, and data governance especially important.

Q84 — B

Responsible AI applies throughout the AI lifecycle because risks can arise during planning, data collection, development, deployment, and ongoing monitoring.

Q85 — B

Political deepfakes can spread misinformation, undermine public trust, and influence democratic processes.

Q86 — B

Overall accuracy can hide unequal performance across demographic groups, making subgroup evaluation essential.

Q87 — A

A system that blocks many legitimate activities while detecting all attacks has a high false positive rate.

Q88 — B

Being transparent about AI limitations encourages appropriate user trust and verification.

Q89 — A

Regular retraining after monitoring performance and fairness is an example of continuous AI governance.

Q90 — B

Improving representation in training data and reassessing fairness helps mitigate historical and representation bias.

Q91 — A

AI Safety addresses unintended harmful outcomes, while AI Security protects AI systems from intentional attacks such as prompt injection.

Q92 — A

Outdated training data can lead to hallucinations or recommendations that no longer align with current knowledge.

Q93 — B

An AI governance committee provides oversight, accountability, and structured decision-making for high-risk AI systems.

Q94 — A

Responsible mental health AI should include escalation mechanisms and human oversight for high-risk situations.

Q95 — B

Responsible AI requires evaluating fairness, robustness, privacy, transparency, explainability, and safety in addition to accuracy.

Q96 — B

Compliance is one aspect of Responsible AI. Organizations often need to exceed legal minimums to meet ethical expectations and build trust.

Q97 — A

Publishing documentation about intended use, assumptions, and limitations improves transparency and helps users understand appropriate use.

Q98 — B

Human-in-the-loop governance ensures AI supports—not replaces—human judgment, particularly for high-impact decisions.

Q99 — B

Organizations should monitor misuse, clarify intended use, educate users, and strengthen governance when AI is used beyond its designed purpose.

Q100 — B

The overarching goal of Responsible AI is to ensure AI systems are trustworthy, ethical, fair, transparent, secure, accountable, and aligned with human values throughout their lifecycle.